CS253 Project IncognIITo

An Anonymous, Interest-Based video chat Platform for IITK community

Team: Code_commanders

THE ORIGIN

- **Problem:**
  IITK's pursuit of excellence can sometimes lead to social isolation and silent pressure. Students need a safe, non-judgmental space for vulnerability.

- **Solution:**
  Our platform acts as a digital sanctuary, bridging this isolation. Through a unified, anonymous ecosystem offering video/text chat and interest-based matchmaking, we empower students to find meaningful peer support and overcome loneliness.

- **Potential Users:**
  Stressed scholar, introverted peer, the newcomer seeking guidance.

Requirements (Functional)

- The website must allow registration through IITK email IDs only and keep a user’s identity anonymous.
- The application should provide users functionality to modify their profile picture, display name and chosen interests.
- The application shall provide the functionality of finding partners with similar interests.
- The application should allow users to see and hear the matched user through live video and audio feed.
- The system shall provide a robust feedback mechanism allowing users to rate and flag interactions.

Requirements (Functional)

- The software should also allow users to connect with each other and chat even after the session ends.
- The system shall provide a comprehensive Admin Console for platform management and moderation.
- Admin should have ability to see all registered users and ban them depending on the reports issued against them
- The application should allow users to reset forgotten passwords via **OTP verification** sent to their registered IITK email IDs

Requirements (Non-Functional)

- **Privacy:** Keep user identity hidden during interaction.
- **Security:** Ensure secure authentication, data protection, and controlled access.
- **Realtime performance:** Fast matchmaking and low-latency chat/video coordination.
- **Reliability:** Duplicate-session protection, persistent chat history.
- **Usability:** Simple onboarding and a clear flow from registration to interaction.
- **Moderation readiness:** Reporting, blocking, and admin action support.

USER LIFECYCLE

01
Register
With IITK email ID.

02
Verify
OTP and set password.

03
Setup
Log in & complete profile.

04
Queue
Join the match-making pool.

05
Match
Get paired into live session.

06
Interact
Chat or connect further.

07
Safety
Report or block if needed.

08
Admin
Review and take action.

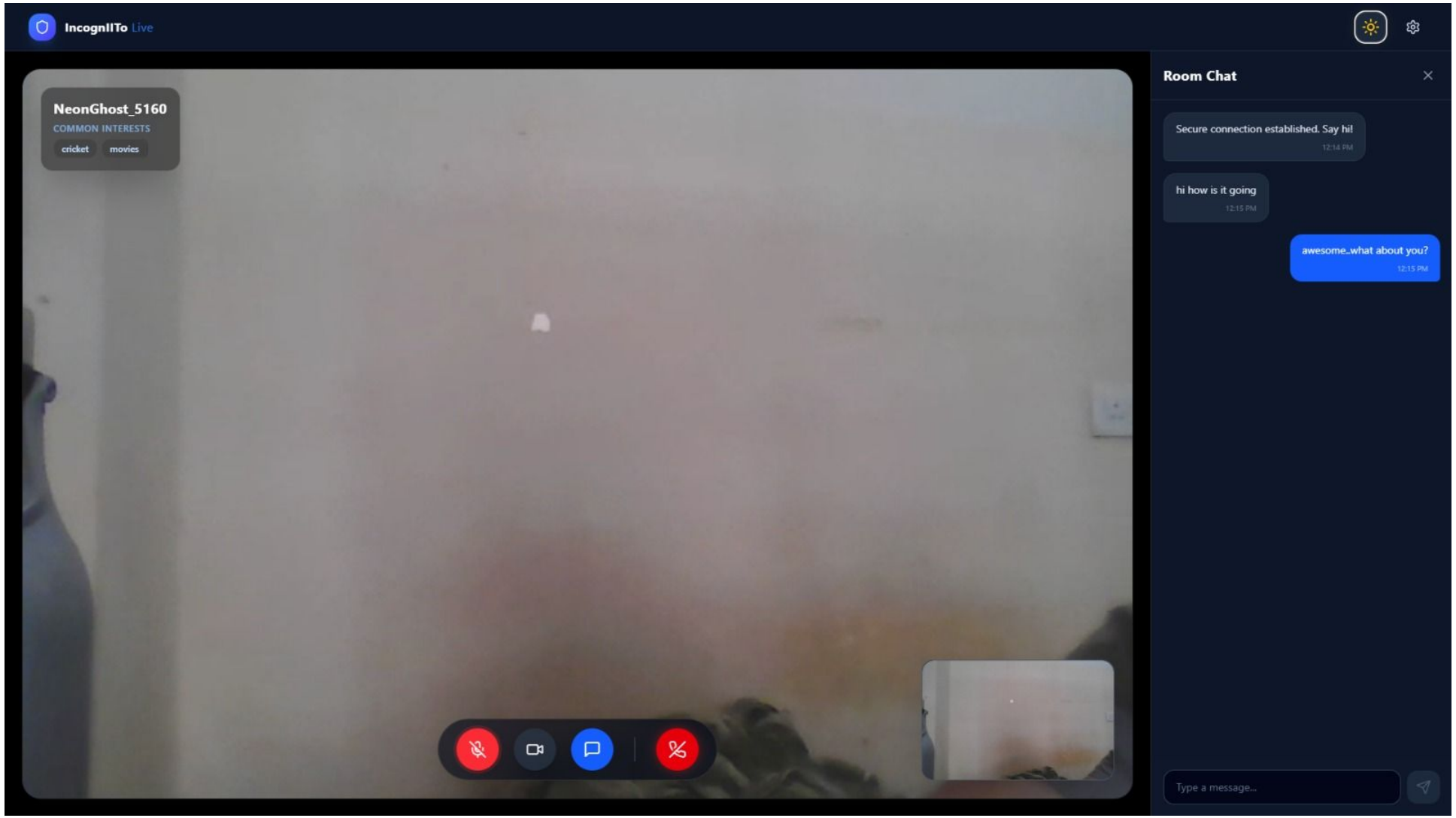
Key features

- Mandatory login via IITK email IDs implemented with Node.js, Express, and secure **JWT/bcrypt** authentication.
- Responsive React/TypeScript UI styled with **Tailwind CSS**, featuring dark/light modes and **Cloudinary-hosted** profile assets.
- A single active session is allowed per user
- Vite utilized as the high-performance build tool to ensure rapid initial load times and a seamless, lag-free landing experience for users.

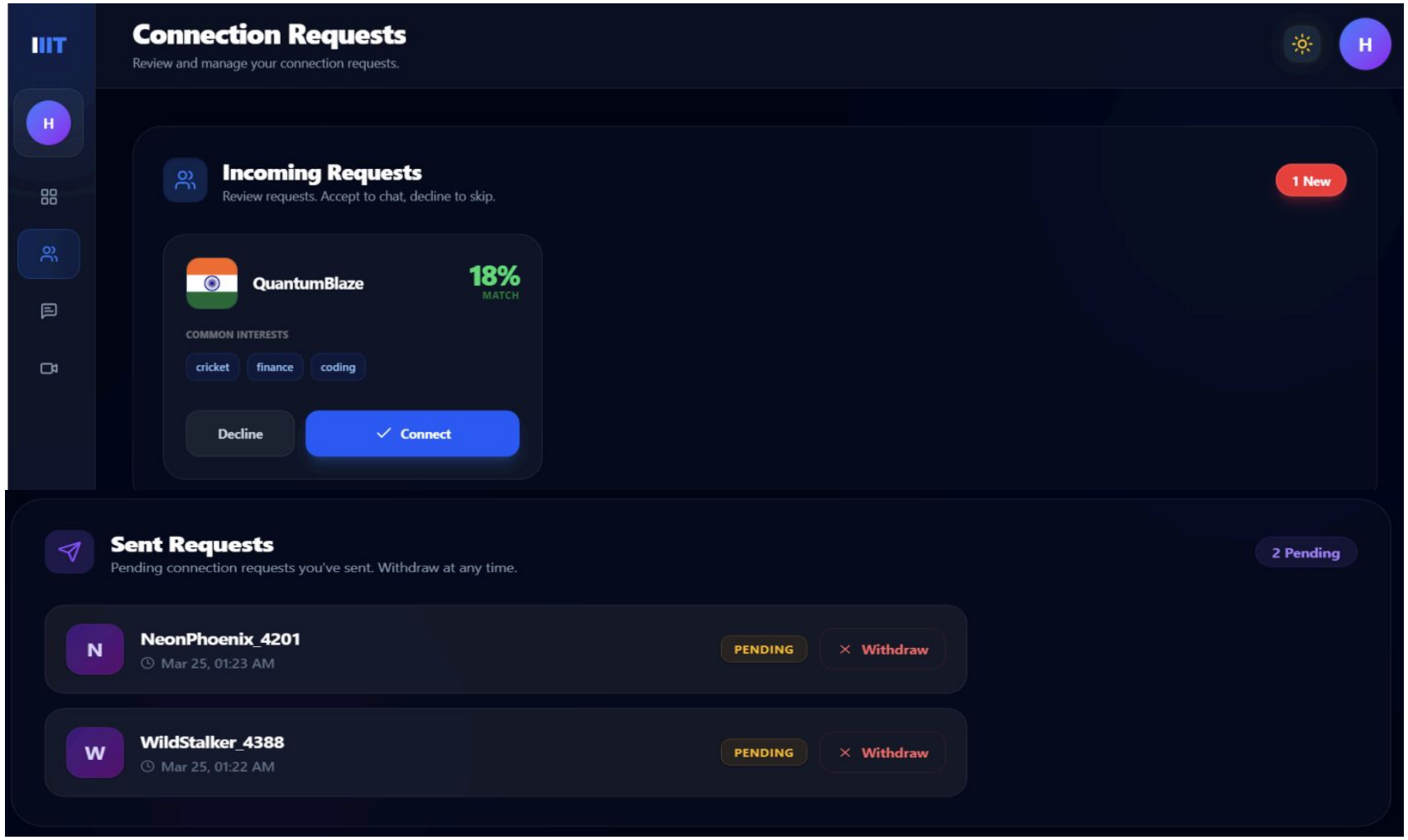
Key features

- A smart matching algorithm that pairs users based on shared academic disciplines, niche hobbies, or other interests keeping idle waiting time minimum.
- The algorithm utilizes the **Jaccard Similarity Coefficient** to quantify the "connection potential" between users
- To balance conversation quality with user patience, the system employs a **decaying threshold logic** that progressively lowers similarity requirements (from 40% down to 0%) as a student's wait time increases
- Redis serves as the high-performance, in-memory queue providing O(1) fast lookups and handles all atomic state transitions to ensure users are matched uniquely and securely without race conditions.

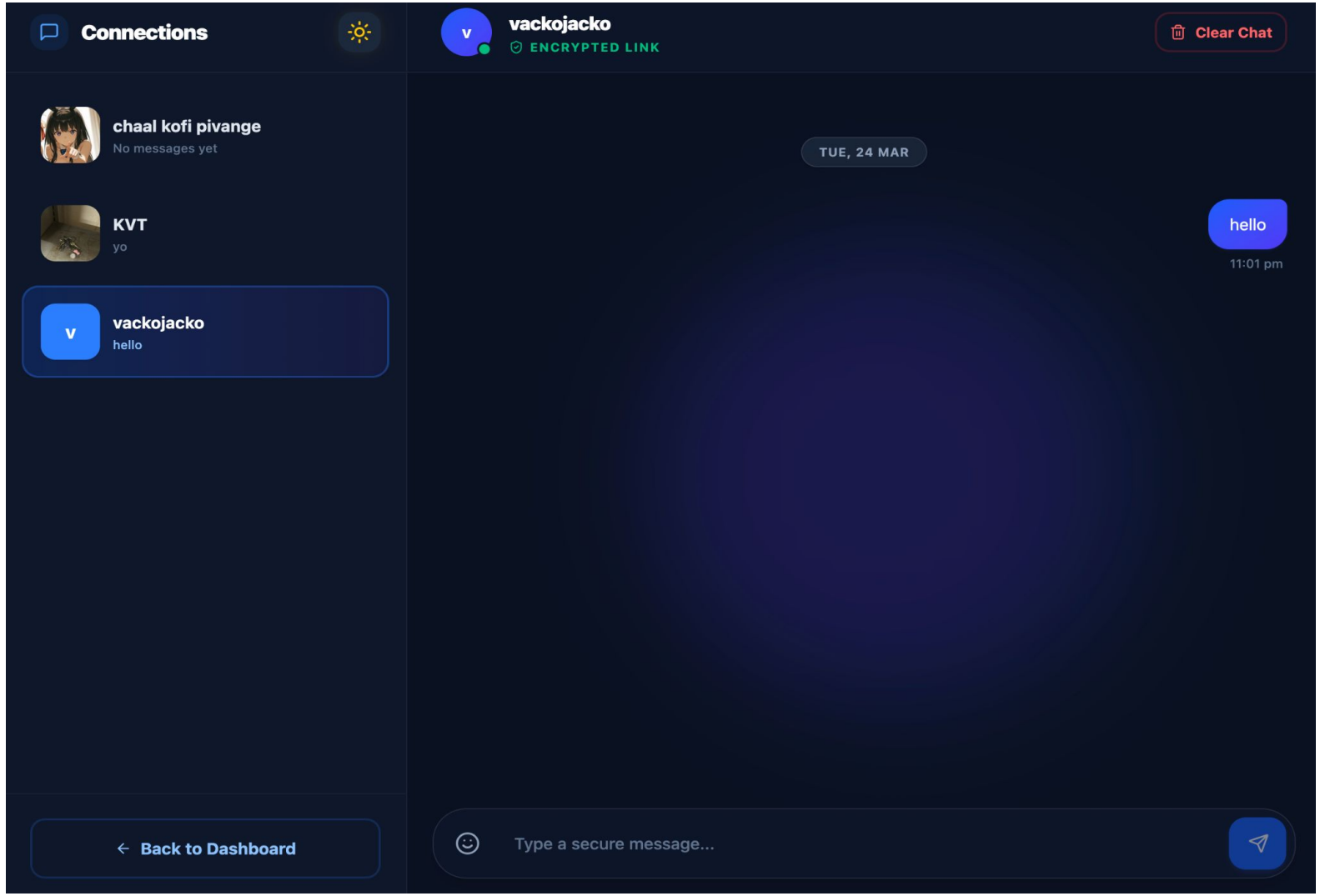
- Seamless peer-to-peer video streaming and real-time text chat powered by **WebRTC** and **Socket.IO** for low-latency connections.



- Mutual-consent connections allows to take the conversation from anonymous to lasting friendship



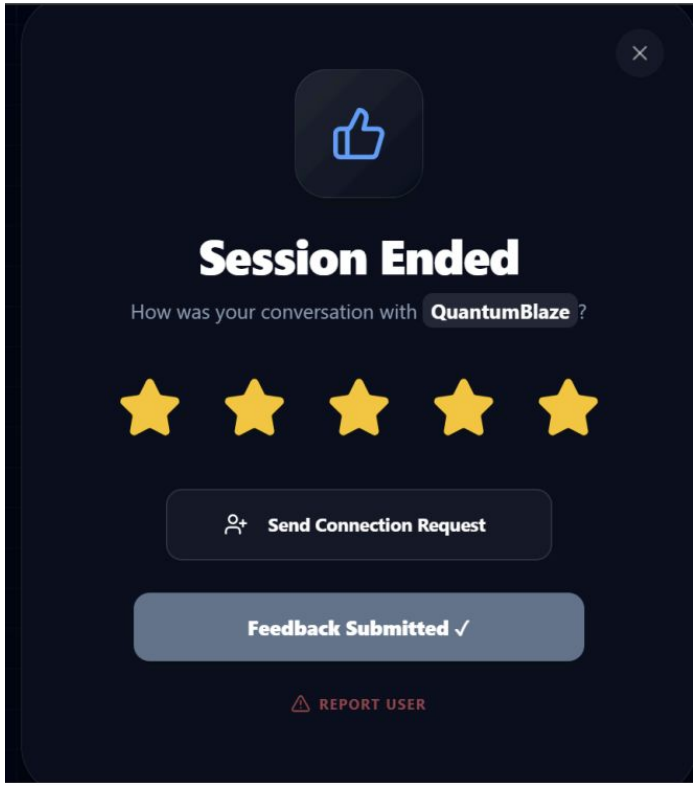
- Chat interface to develop friendships beyond anonymous video calls

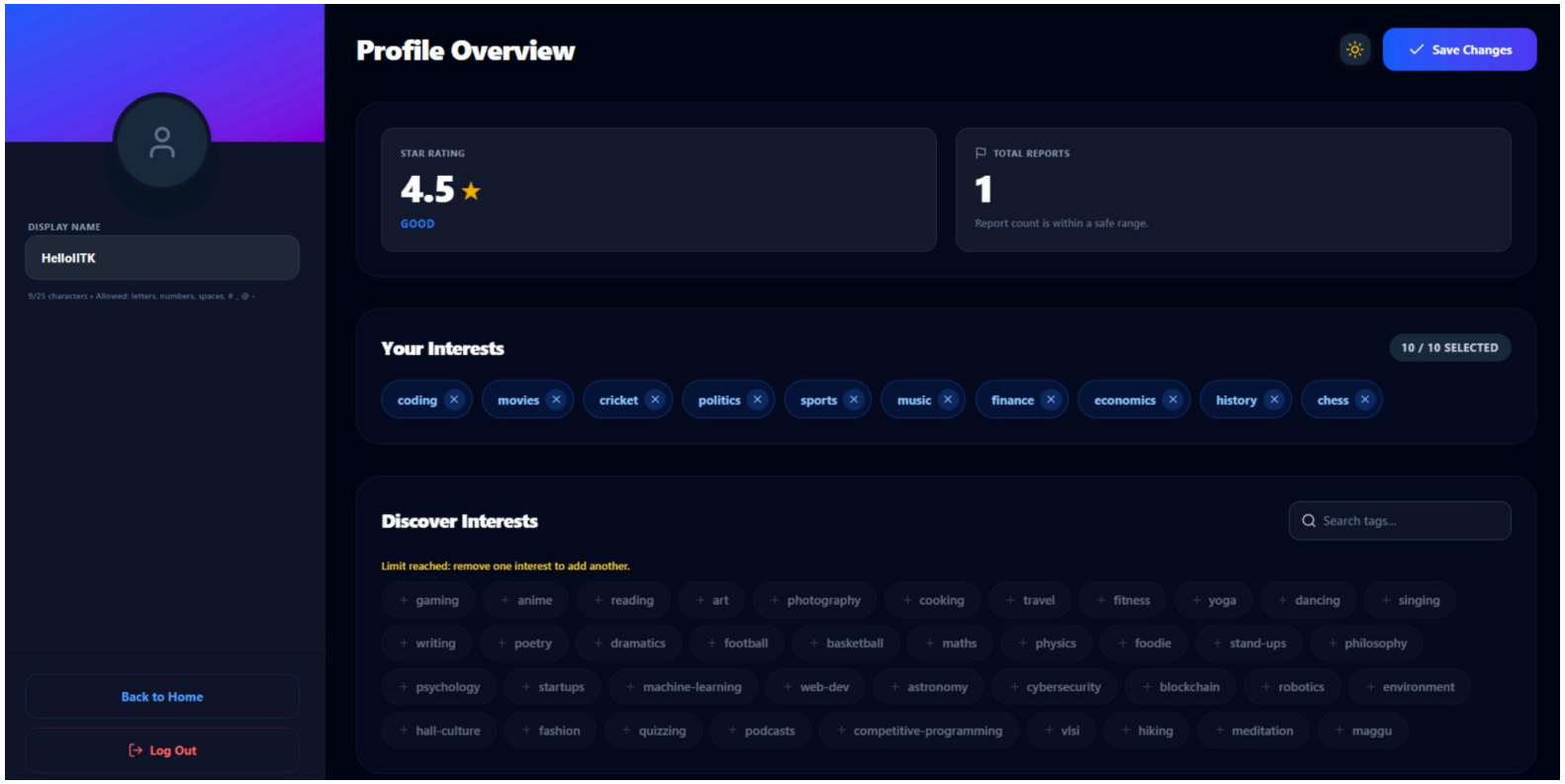


- Encrypted and private interactions.
- Interactive text overlay for accessibility.

- A flexible user profile that allows students to update their "topics of interest" and display name to reflect their evolving needs.

- Post-interaction ratings that reward supportive behavior and prioritize respectful users





- An Admin Console which serves as a centralized governance dashboard for user oversight and report management

- Administrators can ban a user depending on reports issued against him/her.

DEMO TIME

Experience IncognIITo

INCOGNIITO

FUTURE DEVELOPMENT PLANS

Recommendation system

A recommendation system can be incorporated in the matching algorithm to further improve the match quality.

Suggests better matches using past interactions, feedback, and user preferences.

Automated banning

A ML model can be integrated to automate the ban of reported users thereby getting rid of manual intervention
—> Use NLP on chat messages
—> Reduce admin workload
—> Faster response to harmful users

FUTURE DEVELOPMENT PLANS

Adaptive Matchmaking

Dynamic scoring model using session history, feedback, and patterns to improve match quality.

Scalable Infrastructure

Transition to horizontally scalable architecture for concurrent loads beyond the IITK community

Session Analytics

Finally, we plan to incorporate session analytics to track user engagement, match success rates, and usage patterns, enabling us to make data-driven decisions and continuously improve the platform.

LESSONS LEARNT

- **HANDS ON LEARNING:** This is probably the first course where we used and honed our technical skills and imagination to build a practically working website .

- **TEAM SPIRIT:** Managing a project with a team of 10 people for the first time made us realize the importance of teamwork and accepting and respecting others' point of views.

- **PLAN-DRIVEN PROCESS:** The whole project was plan-driven, completing various deadlines on time made us realize the importance of time-management and planning.

QUESTIONS?

We'd love to hear your thoughts on building a more connected IITK.

THANK YOU!

IITK'S EXCLUSIVE NETWORK
IncognIITo
Connect anonymously. Share authentically.
The only video networking platform designed
exclusively for the IIT Kanpur community.
Portal Access
Authenticate to enter the network
Login with Email
Register with Email